Fuzzy Logic and Expert Systems (ET -8309)

Department of Electronics and Telecommunication Engineering
Mbeya University of Science and Technology

Email : dsinkonde2013@gmail

Course Aim

- Many technical disciplines involve environments where all necessary information is not available or is ambiguous. This course will propose fuzzy sets and fuzzy logic as appropriate mathematical models of this incompleteness and ambiguity. The course will focus on example problems in various engineering, mathematics, and science disciplines with an emphasis on environmental issues and processes impacting the environment.

Expected Outcomes

Students can expect to gain the following:

- ✓ Describe basic concept of fuzzy logic and artificial neural network.

- ✓ Explain the mathematical foundation of fuzzy sets, fuzzy logic and fuzzy systems.

- ✓ Design and implement **fuzzy logic controller** for target applications.

Expected Outcomes Contd…

Students can expect to gain the following:

- ✓ Describe learning and adaptation capability of artificial neural networks
- ✓ Perform learning process of various neural networks using standard learning algorithms.
- ✓ Design and implement artificial neural networks for target applications.
- ✓ Use software tools in the simulation and design of intelligent fuzzy logic controller systems.

Assignments and Learning Evaluation

- Mode of Assessment
  - Continuous Assessment (CA) 40%
  - University Examination (UE) 60%

- Number of Credits : 9

- Test to be conducted Starting From Week 7 = 20%

Assignments and Learning Evaluation

- Mode of Assessment
- Continuous Assessment (CA) 40%

Assignments

- Total Score = 10% , Fist of submission on 19th April
- Total Score = 10% , Final of submission on 10th May

- **Tests**
- Test to be conducted Starting From Week 7 = 20%

  - 22nd April, 2024

- University Examination (UE) 60%

- Number of Credits : 9

Assignments

- Project
  a) Analysis of the literature
  b) Design of a solution
  c) Development of the designed solution
  d) Experimental Analysis
  e) Presentation Skills
- Different combinations for a, b, c, d, e but a+b+c+d+e= 20%
- Up to two people
- Knowledge of **MATLAB**
- Topics available at the end of the slide (now :-)

Topics for an Assignments

- **Project**

1. Design and development of Fuzzy Logic for prediction of the Path/sidewalk. A Case Study MUST.

2. Design position tracking and motion prediction using Fuzzy Logic.

3. Design and development Rainfall Prediction Model Improvement by Fuzzy Set Theory

Topics for an Assignments

- **Project**

4. Fuzzy association rule mining and classification for the prediction of UTI in MUST Area.

5. Design the traditional of ER Data model and Extend to fuzzy ER data model.

6. Fuzzy association rule mining and classification for the prediction of maternal deaths in Mbeya City context.

Topic to be Covered

Introduction:

Introduction to fuzzy logic, neuro fuzzy, and soft computing, from conventional AI to computational intelligence, Neural Network, Evolutionary computation, Neuro fuzzy and soft computing characteristics, *Fuzzy set theory; Basic definition & terminology, set theoretic operations, MF formulation & parameterization, Fuzzy union, intersection & compliment.*

Topic to be Covered

Fuzzy Rules & fuzzy Reasoning: Extension Principles and Fuzzy Relations, Fuzzy if then rules, linguistic Variables, Fuzzy reasoning, compositional rules of inference, Fuzzy systems as function estimators, Fuzziness as multivalence.

Fuzzy Inference system: Mamdani Fuzzy models, other variants, Sugeno Fuzzy models, Tsukamoto Fuzzy models, Input space partitioning, Fuzzy modeling, intelligent behaviour as adaptive model free estimation.

Topic to be Covered

Adaptive Fuzzy Systems: Fuzzy sets and systems, Fuzziness in a probabalistic world, randomness V/S ambiguity, The universe as a fuzzy set, The geometry of fuzzy sets, the Fuzzy entropy theorem, The subsethood theorem, the entropy subsethood theorem, Introduction to Fuzzy associative memories.

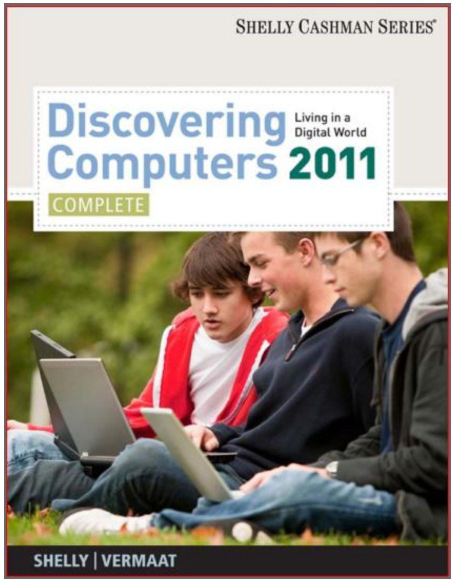
Regression and optimization: Introduction to least-square methods for system identification and derivative based optimization, derivative free optimization: Genetic algorithm, Simulated Annealing, Random search, Downhill simplex search.

General Course Information

Textbooks/References

1. S.R. Jang, Sun & Mizutani, "Neuro-Fuzzy and soft computing," Prentice Hall Inc, 1997.
2. Introduction of artificial neural systems J.M.ZURADA Jaico publication House 1997
3. Neural networks : comprehensive foundation S.IIAYKIN McMillian College Publishing company inc. 1994
4. Neuro control and its application S.OMATU, M.KIIALID, R.YUSOF. Spring Verlag London Ltd. 1996.
5. B. Kosko, "Neural Network & Fuzzy Systems," PHI pvt ltd.
6. Haykin, "Fuzzy Logic & Artificial Neural Network: A Comprehensive Foundation," Asea

Lecture 1:



Introduction to Fuzzy set theory

Topic to be covered

- *A bit of history*
- *Introduction to Neuro-fuzzy and Soft computing*
- *Brief Review of convention Sets*
- *Introduction of Fuzzy sets*
- *Membership Functions.*
- *Operations on Fuzzy Sets*

Domain Included in Artificial Intelligence

- *Artificial intelligence is divided into several domains that work in specific fields.*
- *Here are few of these domains:*

1. *Fuzzy Logic System*
2. *Expert Systems*
3. *Robotics*
4. *Neural Networks*
5. *Natural Language Processing*

What is computing?

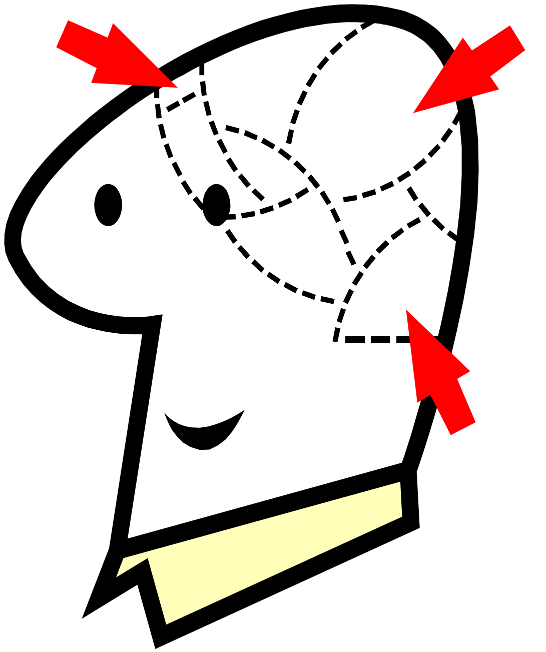
- Counting, calculating
- *The discipline of computing is the systematic study of algorithmic* processes that describe and transform information: their theory, analysis, design, efficiency, implementation, and application.
- Types of computing
  - Hard computing
  - Soft Computing

Differences between hard and soft computing

Hard Computing	Soft computing
Precisely stated analytical model	Tolerant to imprecision, uncertainty, partial truth, approximation
Based on binary logic, crisp systems, numerical analysis, crisp software	Fuzzy logic, neural nets, probabilistic reasoning.
Programs are to be written	Evolve their own programs
Two values logic	Multi valued logic
Exact input data	Ambiguous and noisy data
Strictly sequential	Parallel computations
Precise answers	Approximate answers

- Essence of SC:-
  - Accommodation with the pervasive imprecision of the real world

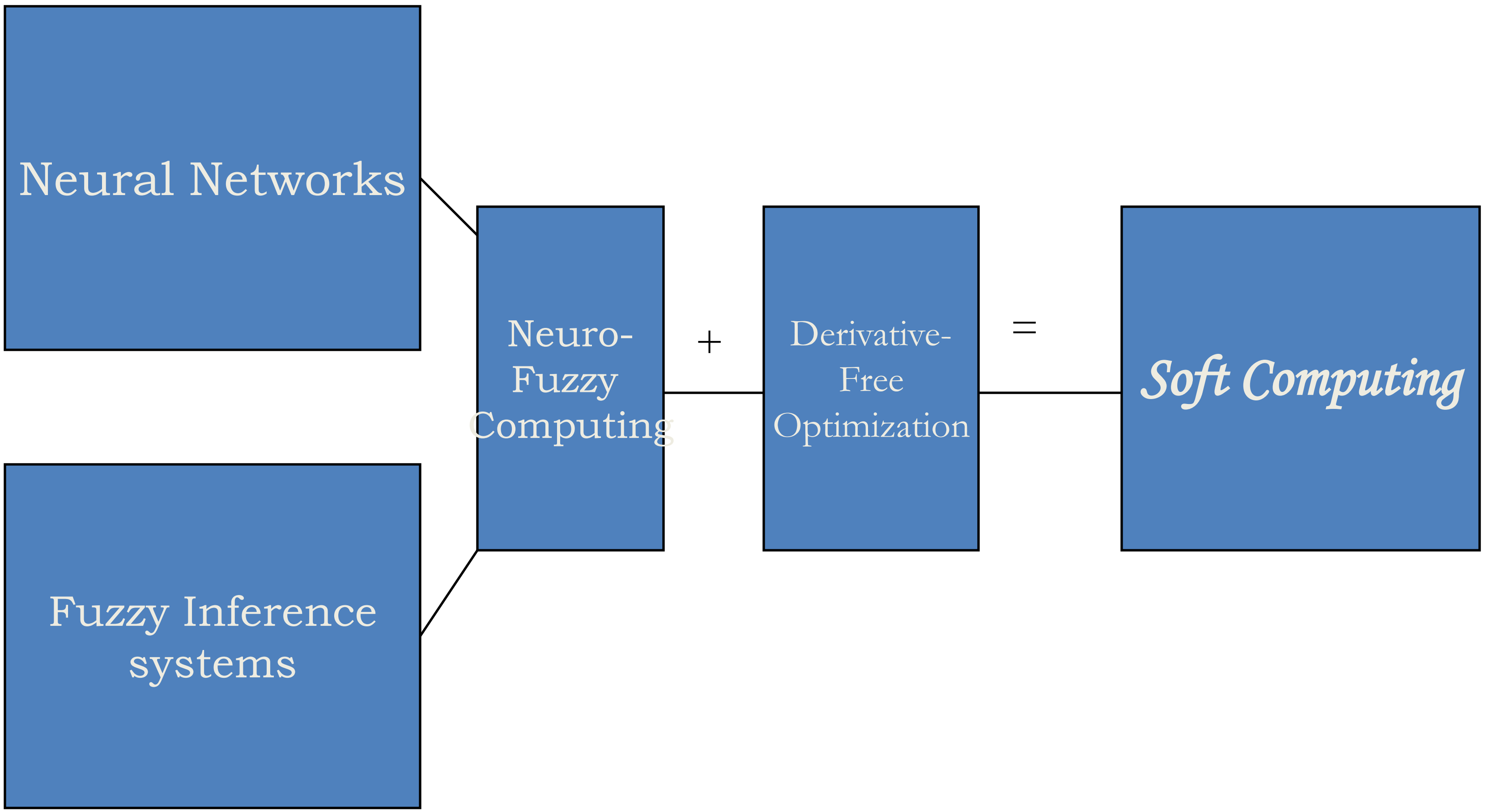
- Principle of SC:-
  - Exploit uncertainty to achieve robustness and better rapport with reality

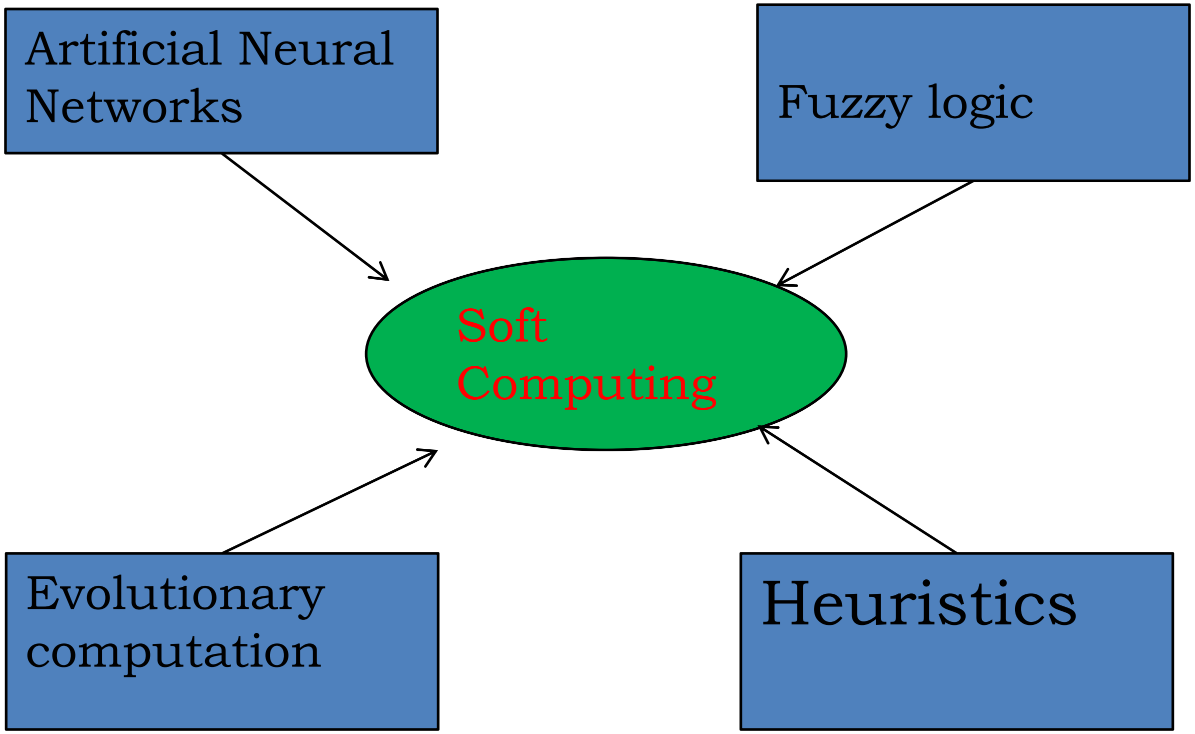


Artificial intelligence

- If intelligence can be induced in machines it is called as artificial intelligence.
- Soft computing is a part of artificial intelligent techniques
- Closed related to machine intelligence/computational intelligence

What is Soft computing




Artificial Neural Networks
Fuzzy logic
Soft Computing
Evolutionary computation
Heuristics

Introduction SC

- SC is an innovative approach to constructing computationally intelligent systems

- Intelligent systems that possess human like expertise within a specific domain, adapt themselves and learn to perform better in changing environments

- These systems explain how they make decisions or take actions

- They are composed of two features: “adaptivity” & “knowledge

Introduction Contd….

- Neural Networks (NN) that recognize patterns & adapts themselves to cope with changing environments

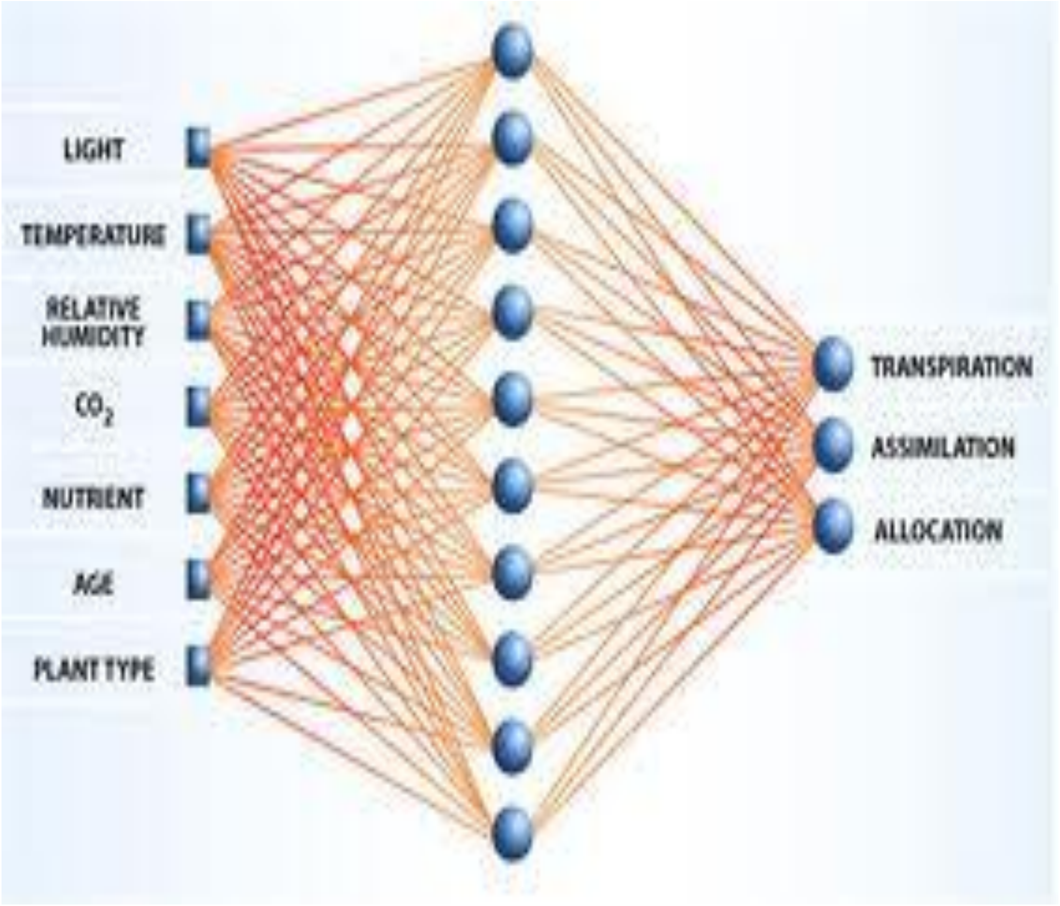
- Fuzzy inference systems that incorporate human knowledge & perform inference & decision making

Adaptivity + Expertise = NF & SC

Neural Network (NN)

- Imitation of the natural intelligence of the brain
- Parallel processing with incomplete information
- Nerve cells function about 106 times slower than electronic circuit gates, but human brains process visual and auditory information much faster than modern computers
- The brain is modeled as a continuous-time non linear dynamic system in connectionist architectures
- Connectionism replaced symbolically structured representations
- Distributed representation in the form of weights between a massive set of interconnected neurons

Neural Network (NN)



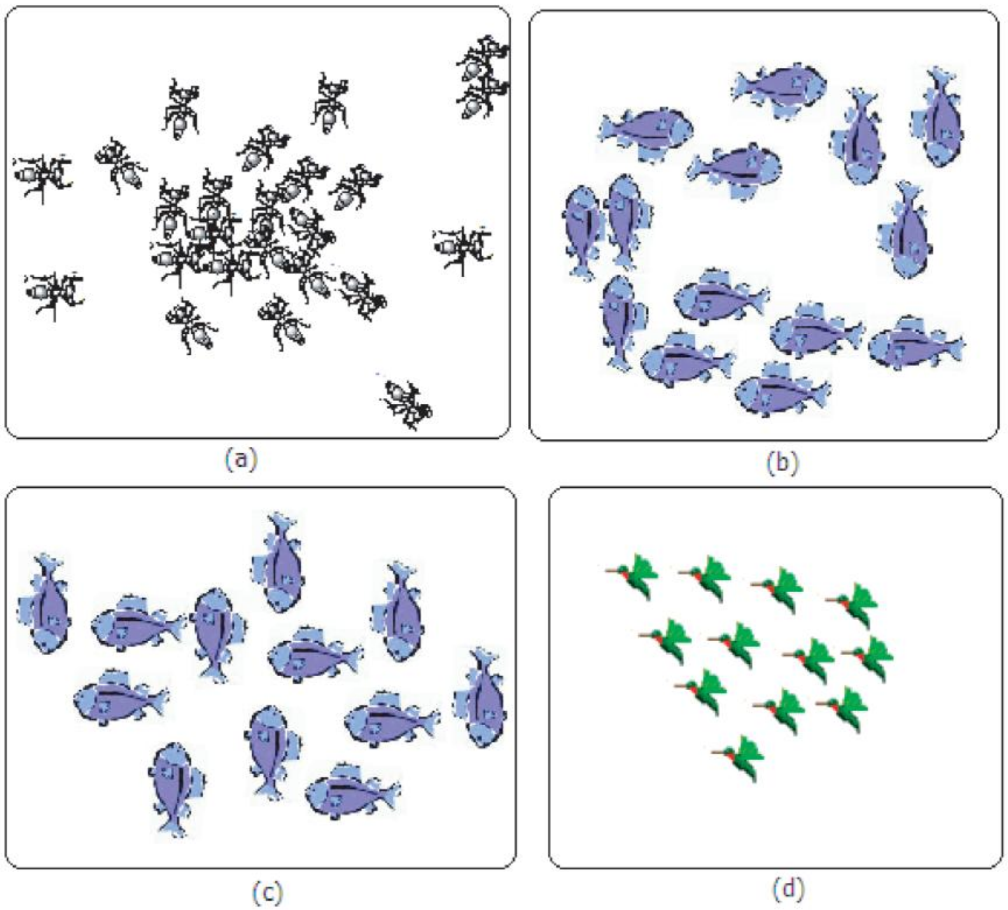
What is the difference between Fuzzy Logic and Neural Networks?

- Fuzzy logic allows making definite decisions based on imprecise or ambiguous data.
- ANN tries to incorporate human thinking process to solve problems without mathematically modeling them.
- Both these methods can be used to solve nonlinear problems, and problems that are not properly specified, but they are not related.
- ANN tries to apply the thinking process in the human brain to solve problems.

Latest developments in the field of soft computing

- Areas of image processing
  - Image retrieval
  - Image analysis
- Remote sensing
- Data mining
  - Swarm intelligence
  - Diffusion process
  - Agent’s technology

Swarm Technology



(a) (b) (c) (d)

SC Constituents and Conventional AI

- "SC is an emerging approach to computing which parallel the remarkable ability of the human mind to reason and learn in a environment of uncertainty and imprecision" [Lotfi A. Zadeh, 1992]
- SC consists of several computing paradigms including:
  - NN
  - Fuzzy set theory
  - Approximate reasoning
  - Derivative-free optimization methods such as genetic algorithms (GA) & simulated annealing (SA)

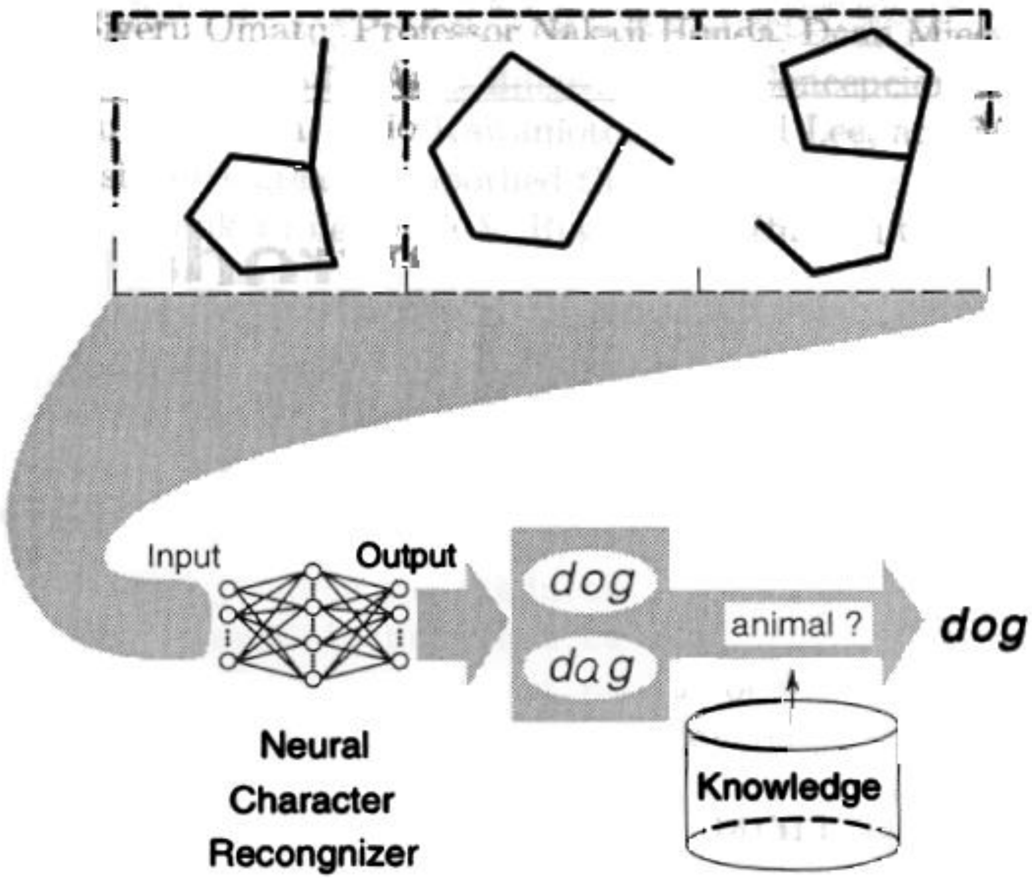
SC constituents (the first three items) and conventional AI

Methodology	Strength
Neural network	Learning and adaptation
Fuzzy set theory	Knowledge representation via fuzzy if-then rules
Genetic algorithm and simulated annealing	Systematic random search
Conventional AI	Symbolic manipulation

Contd….

- These methodologies form the core of SC
- In general, SC does not perform much symbolic manipulation
- SC in this sense complements conventional AI approaches

Character recognizer



Features of Conventional AI

- From conventional AI to computational intelligence
  - Conventional AI manipulates symbols on the assumption that human intelligence behavior can be stored in symbolically structured knowledge bases: this is known as: “ The physical symbol system hypothesis
  - The knowledge-based system (or expert system) is an example of the most successful conventional AI product

NF and SC characteristics

- With NF modeling as a backbone, SC can be characterized as:
  - Human expertise (fuzzy if-then rules)
  - Biologically inspired computing models (NN)
  - New optimization techniques (GA, SA, RA)
  - Numerical computation (no symbolic AI so far, only numerical)

NF and SC Characteristics Contd…

- New application domains: mostly computation intensive like adaptive signal processing, adaptive control, nonlinear system identification etc
- Model free learning:-models are constructed based on the target system only
- Intensive computation: based more on computation

NF and SC Characteristics Contd…

- Fault tolerance: deletion of a neuron or a rule does not destroy the system. The system performs with lesser quality

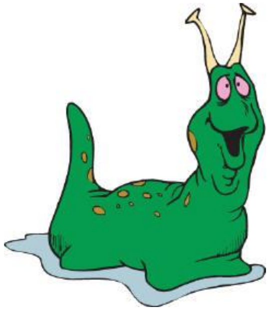
- Goal driven characteristics:- only the goal is important and not the path.

- Real world application:- large scale, uncertainties

WHAT IS FUZZY LOGIC?

- Definition of fuzzy
  - Fuzzy – "not clear, distinct, or precise; blurred"
- Definition of fuzzy logic
  - A form of knowledge representation suitable for notions that cannot be defined precisely, but which depend upon their contexts.

TRADITIONAL REPRESENTATION OF LOGIC



Slow

Fast

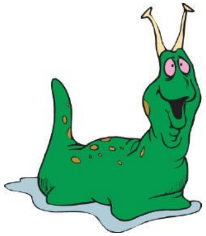
Speed = 0

Speed = 1

```
bool speed;
get the speed
if ( speed == 0) {
//  speed is slow
}
else {
//  speed is fast
}
```

FUZZY LOGIC REPRESENTATION

- For every problem must represent in terms of fuzzy sets.
- What are fuzzy sets?



Slowest

[0.0 – 0.25]

Slow

[0.25 – 0.50]

Fast

[0.50 – 0.75]

Fastest

[0.75 – 1.00]

FUZZY LOGIC REPRESENTATION CONT.

Slowest Slow Fast Fastest

```
float speed;
get the speed
if ((speed >= 0.0)&&(speed < 0.25)) {
//  speed is slowest
}
else if ((speed >= 0.25)&&(speed < 0.5))
{
//  speed is slow
}
else if ((speed >= 0.5)&&(speed < 0.75))
{
//  speed is fast
}
else // speed >= 0.75 && speed < 1.0
{
//  speed is fastest
}
```

ORIGINS OF FUZZY LOGIC

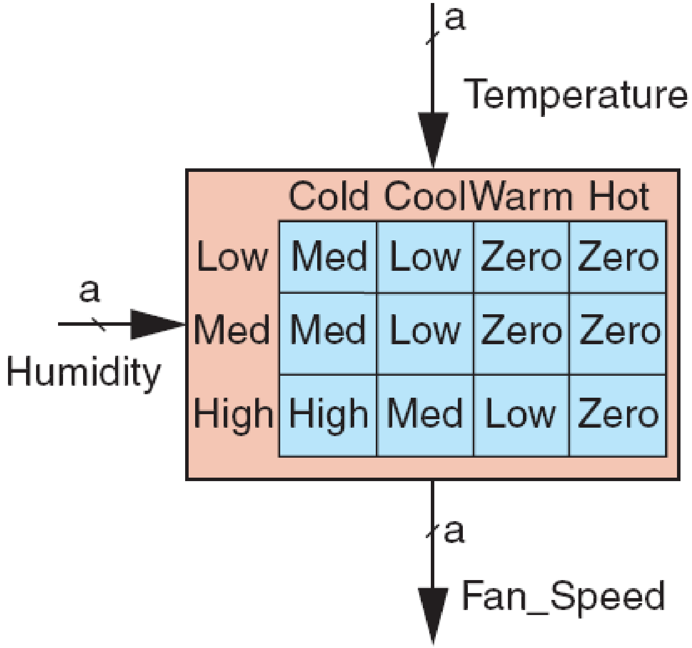
- Traces back to Ancient Greece
- Lotfi Asker Zadeh (1965)
  - First to publish ideas of fuzzy logic.
- Professor Toshire Terano (1972)
  - Organized the world's first working group on fuzzy systems.
- F.L. Smidth & Co. (1980)
  - First to market fuzzy expert systems.

FUZZY LOGIC IN CONTROL SYSTEMS

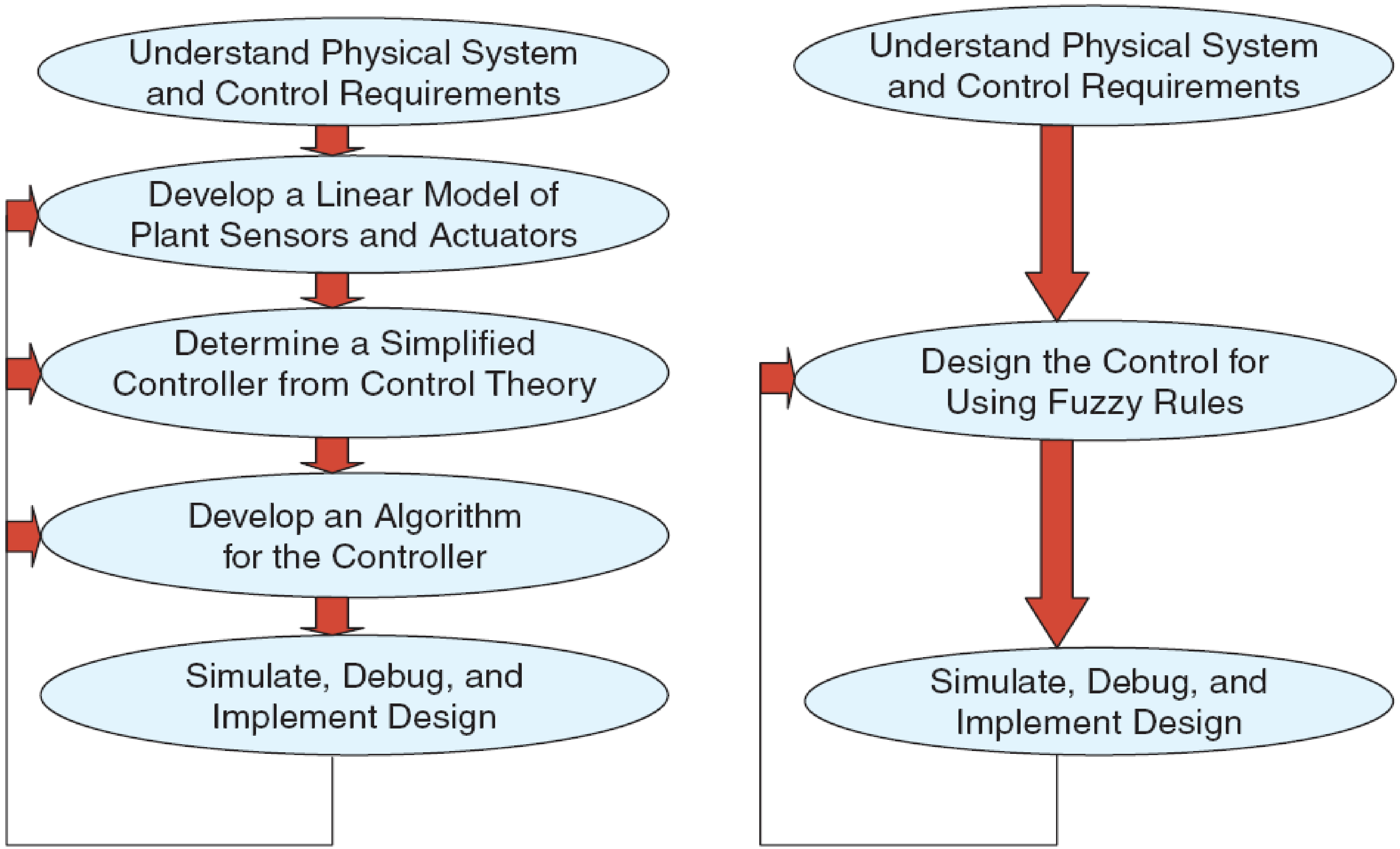
- Fuzzy Logic provides a more efficient and resourceful way to solve Control Systems.
- Some Examples
  - Temperature Controller
  - Anti – Lock Break System (ABS)

TEMPERATURE CONTROLLER

- The problem
  - Change the speed of a heater fan, based off the room temperature and humidity.
- A temperature control system has four settings
  - Cold, Cool, Warm, and Hot
- Humidity can be defined by:
  - Low, Medium, and High
- Using this we can define the fuzzy set.



BENEFITS OF USING FUZZY LOGIC



ANTI LOCK BREAK SYSTEM (ABS)

- Nonlinear and dynamic in nature
- Inputs for Intel Fuzzy ABS are derived from
  - Brake
  - 4 WD
  - Feedback
  - Wheel speed
  - Ignition
- Outputs
  - Pulsewidth
  - Error lamp

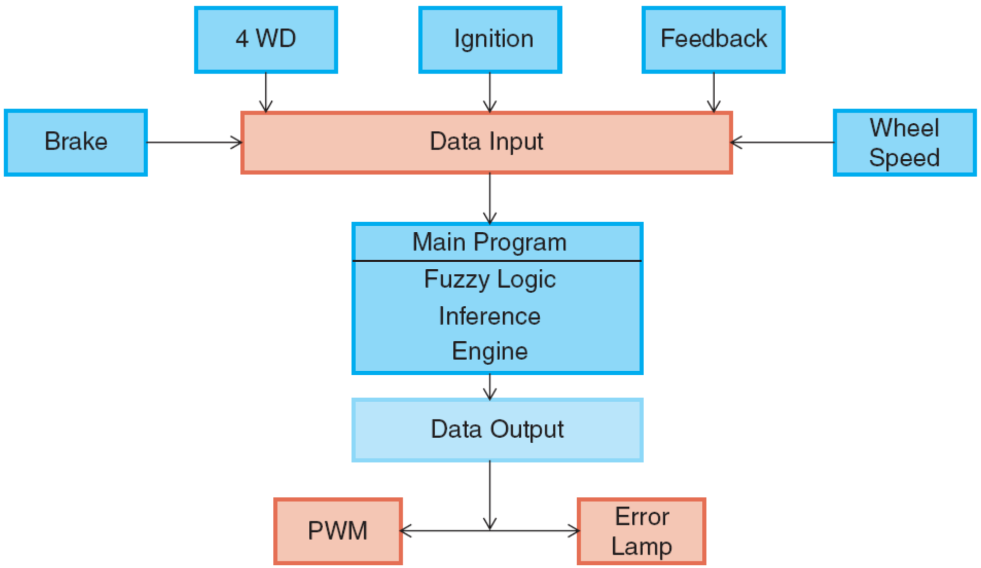


Fig. 6 ABS block diagram

FUZZY LOGIC IN OTHER FIELDS

- Business
- Hybrid Modeling
- Expert Systems

Summary

- Fuzzy logic provides an alternative way to represent linguistic and subjective attributes of the real world in computing.
- It is able to be applied to control systems and other applications in order to improve the efficiency and simplicity of the design process.
- SC is evolving rapidly
- New techniques and applications are constantly being proposed

Fuzzy set theory

- Human brains interpret imprecise and incomplete sensory information provided by perceptive organs
- Fuzzy set theory provides a systematic calculus to deal with such information linguistically
- It performs numerical computation by using linguistic labels stimulated by membership functions
- It lacks the adaptability to deal with changing external environments ==> incorporate NN learning concepts in fuzzy inference systems: NF modeling

Evolutionary computation

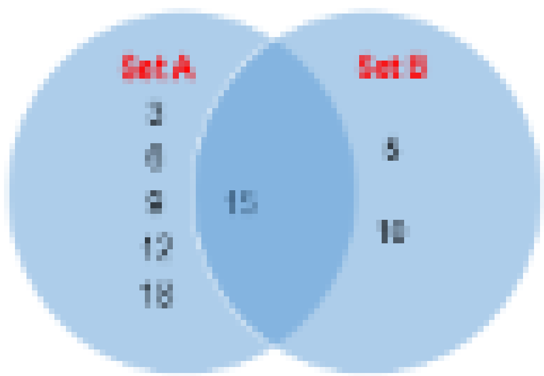
- Natural intelligence is the product of millions of years of biological evolution
- Simulation of complex biological evolutionary processes
- GA is one computing technique that uses an evolution based on natural selection
- Immune modeling and artificial life are similar disciplines based on chemical and physical laws
- GA and SA population-based systematic random search (RA) techniques

Brief Review of convention Sets

- Definition:

  -Sets: A collection of objects having one or more common characteristics.

- Members/Elements: Objects belonging to a set is represented as $A \in P_1$ where P is a set



Brief Review of convention Sets

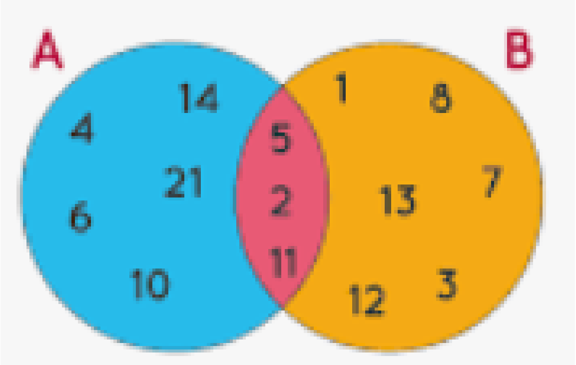
Subset: BP_2 is said to be a subset of set AP_1 if only if $B \in P_2 \Longrightarrow B \in P_1, \forall_B$. This sounds as $B \subseteq A$.

Proper Subset: A set BP_2 could be a proper subset of set AP_1 if only and only if P_2 span is a subset of P_1 and $\exists_A \in P_1$, such that $A \notin P_2$, so then, is written as $B \subset A$.

Sets number operations

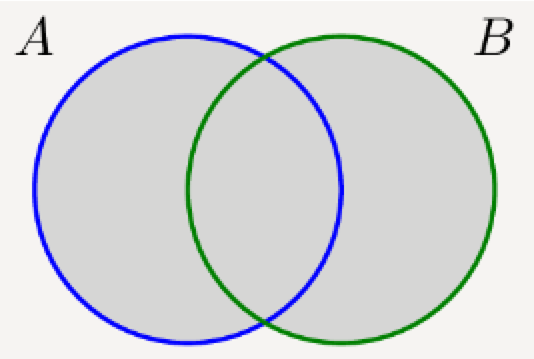
Equal Sets: Two sets A and B are said to be equal $iff\ \forall_x \epsilon A\ and\ \forall_y \epsilon B,\ x = y$.

Intersection Operation: For any two sets A and B if $\exists_x$ common in both A and B, then $x \in (A \cap B)$ where $\cap$ denotes the logical intersection operation.

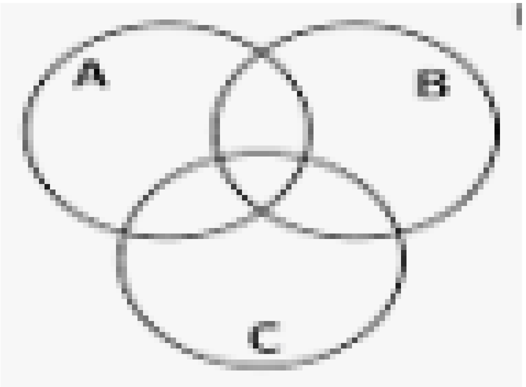


Sets number operations

Union Operation: For any two sets A and B if $\exists_x$ which is a member of either A or B then $x \in (A \cup B)$ where $\cup$ denotes the logical union operation.



Universal Set: A universal set U is a set that has all possible numbers of a particular domain.



fuzzy sets Vs ***Classical set***

Every member x of a fuzzy set, A, is assigned a fuzzy index $\mu_A(x)$ in the universal of (0,1) which is often called as the grade of membership of x in A.

In classical set, membership grade $\mu_A(x)$ is either 0 or 1.

Fuzzy sets: A is a set of ordered pairs given by

$A = ([x, \mu_A(x)] : x \epsilon X)$ where X is universal set and $\mu_A(x)$ is the grade of membership of the object x in A.

usually $\mu_x(x)$ lies in (0, 1).

Membership Functions: A membership function $\mu_x(x)$ is characterized by, $\mu_x : x \rightarrow [0,1],\ x \in X$

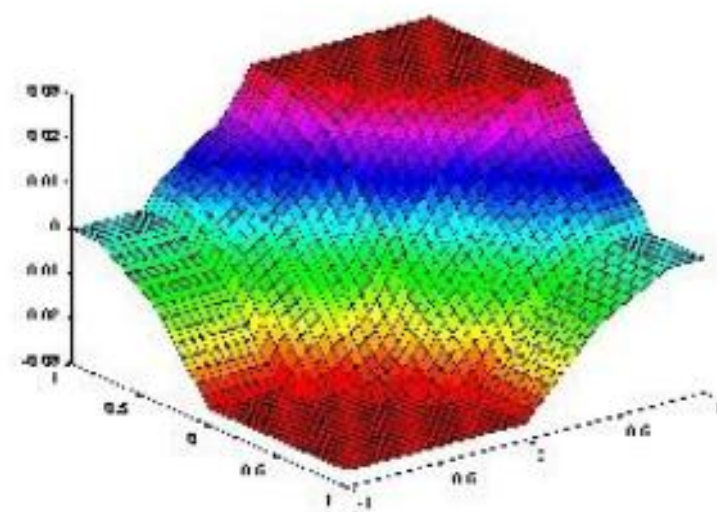
Where x is a real number describing on object or its attribute, X , is the universal of discourse and ACX.

Main characteristics

Fuzzy sets:
precise model in a finite number of points, smooth transition (approximation) among them.

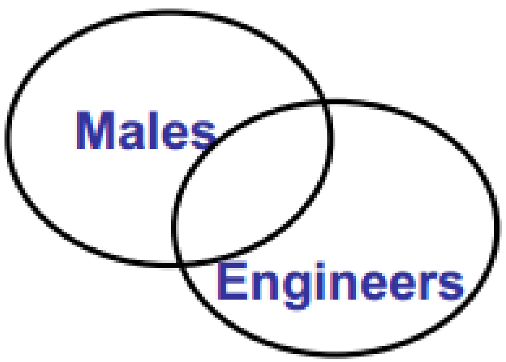
E.g.: **control of a power plant**

We can define what to do at the regimen (e.g., steam temperature =120°, steam pressure 3 atm), and when in critical situations (e.g., steam temperature= 100°), and design a model that smoothly goes from one point to the other.

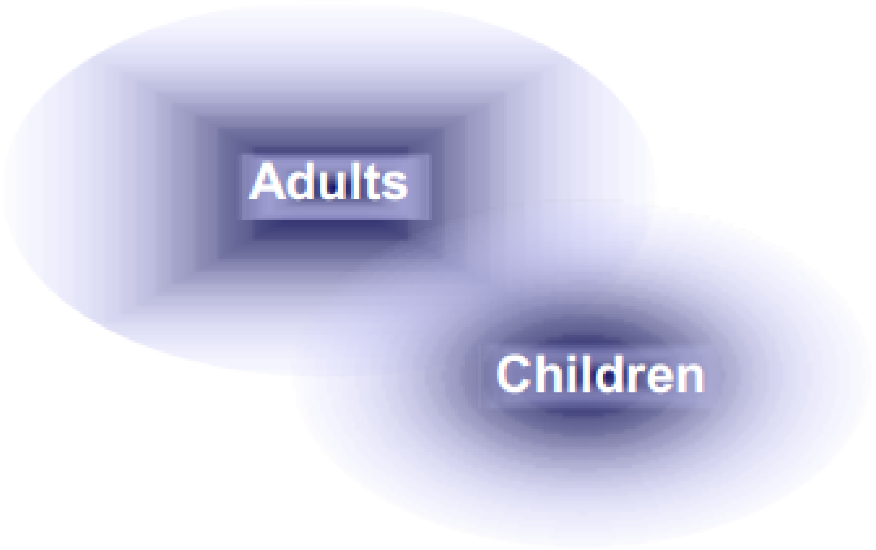


What is a fuzzy set?

A fuzzy set is a set whose membership function may range on the interval [0,1].



Crisp sets



Fuzzy sets

Fuzzy membership functions

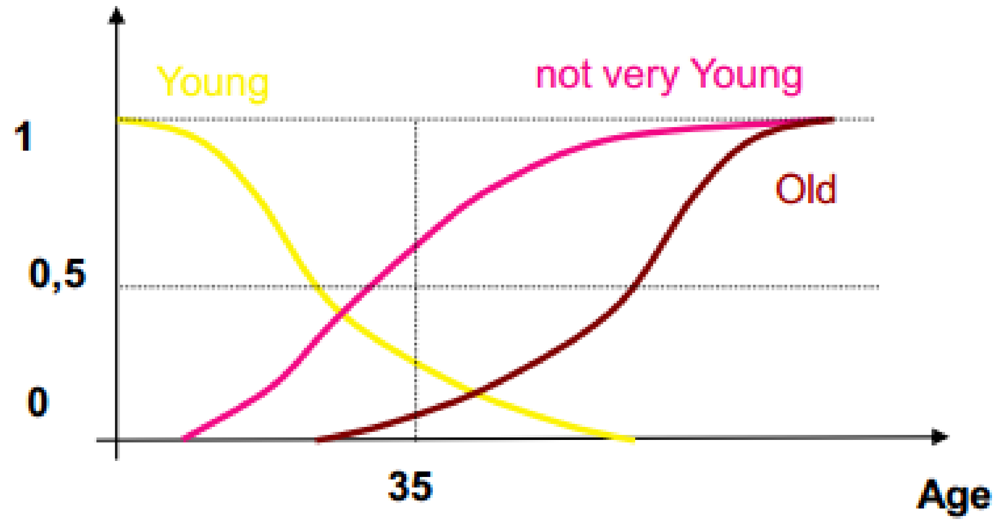
A membership function defines a set

Defines the degree of membership of an element to the set

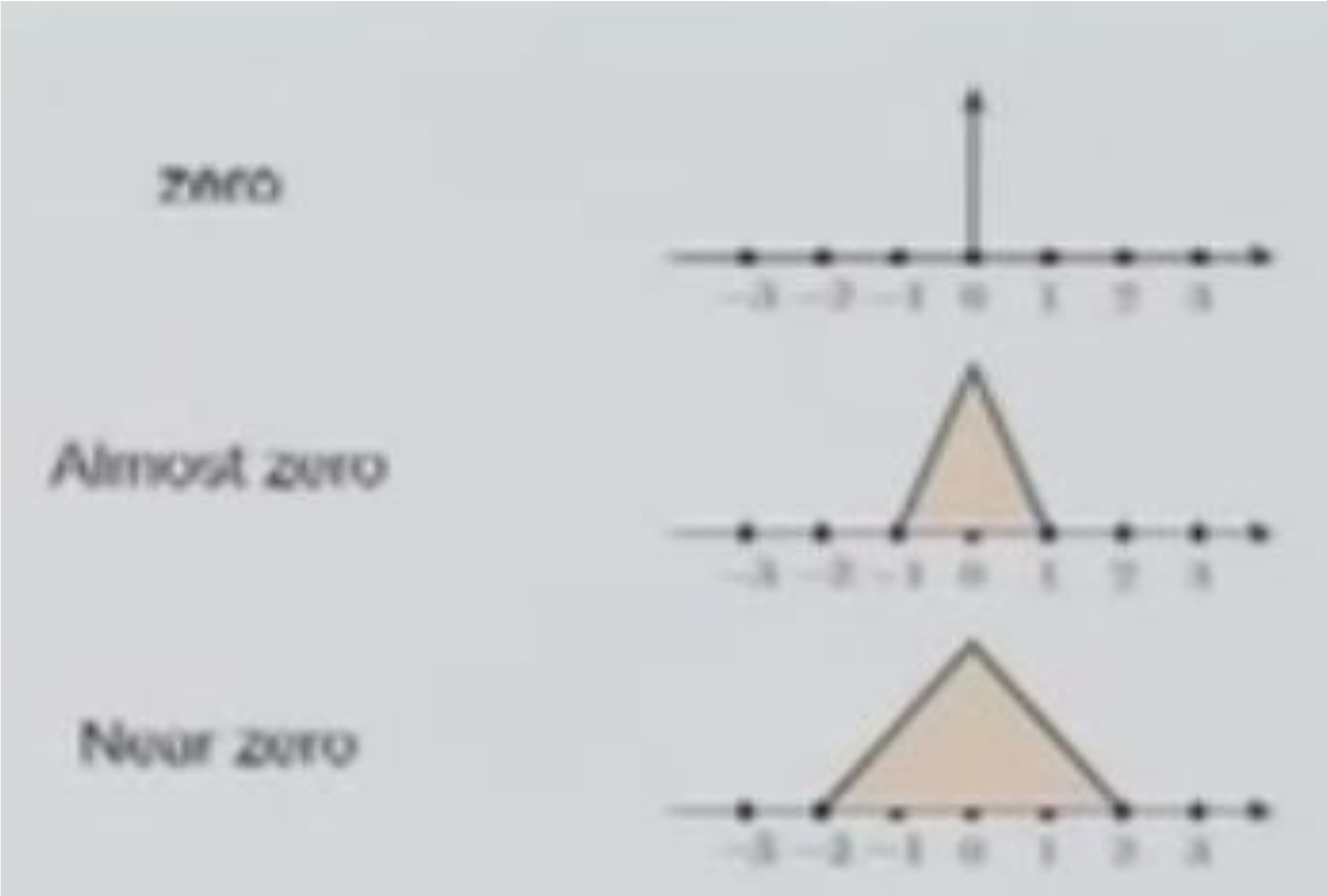
μ: U → [0, 1]

A 35 years old person is:

- Young **with membership 0,3**
- Old **with membership 0,2**
- not very Young **with membership 0,6**



Concept of a Fuzzy membership

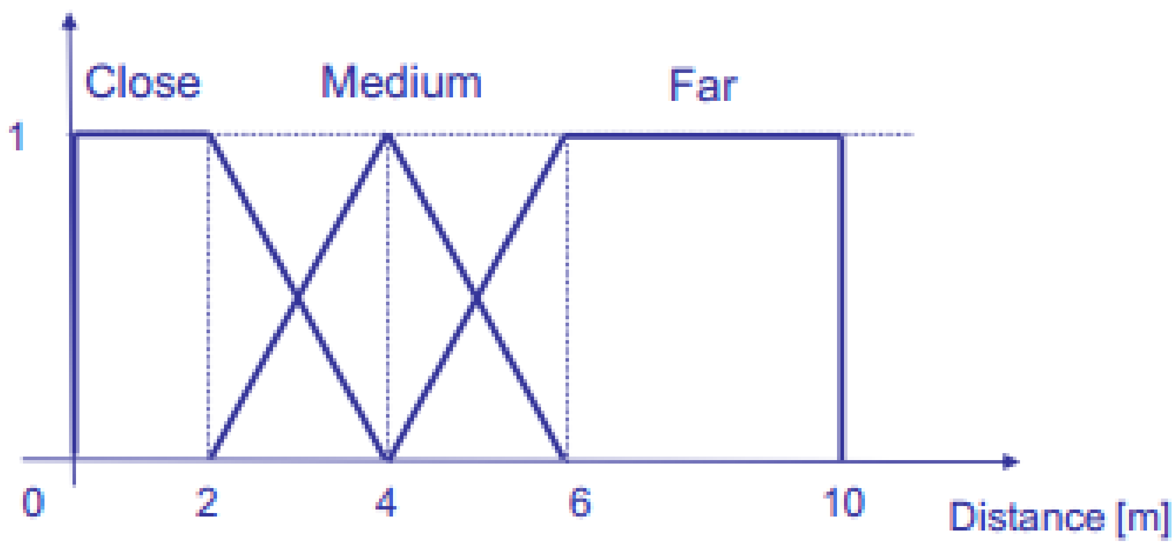


How to define MFs

1. Select a variable
2. Define the range of the variable
3. Identify labels
4. For each label identify characteristic points
5. Identify function shapes
6. Check

Let's try to define some MFs

First of all, the variable...	→	Distance
Range of the variable	→	[0..10]
Labels	→	Close, Medium, Far
Characteristic points	→	0, max, middle values, where MF=1, ...
Function shape	→	Linear



MFs and concepts

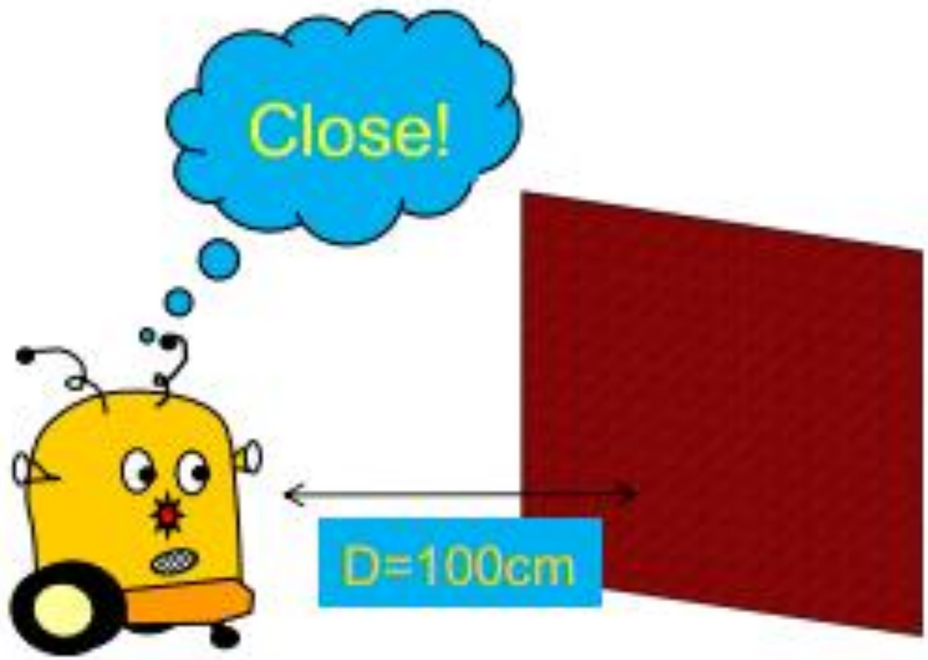
MFs define **fuzzy sets**

Labels denote **fuzzy sets**

Fuzzy sets can be considered as conceptual **representations**

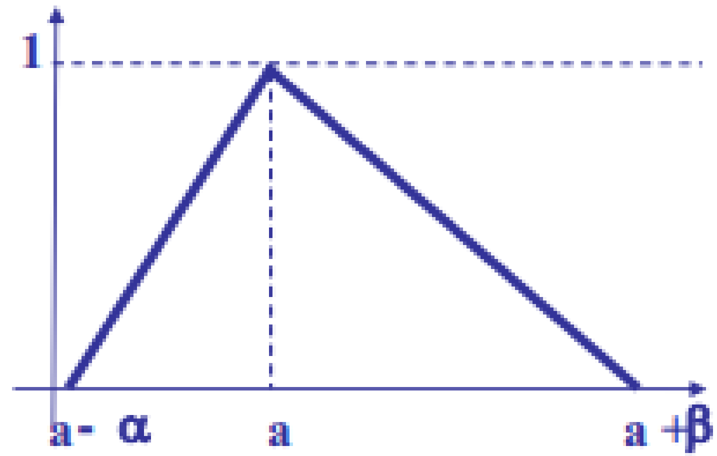
Symbol grounding**:**

reason in terms of concepts and ground them on objective reality

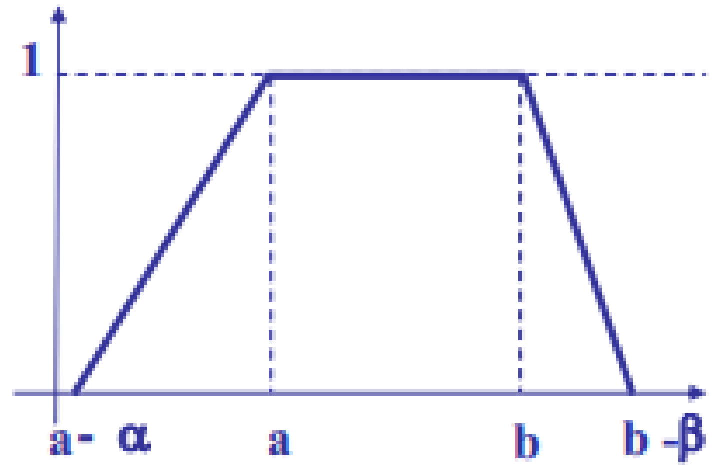


Some conceptual differences

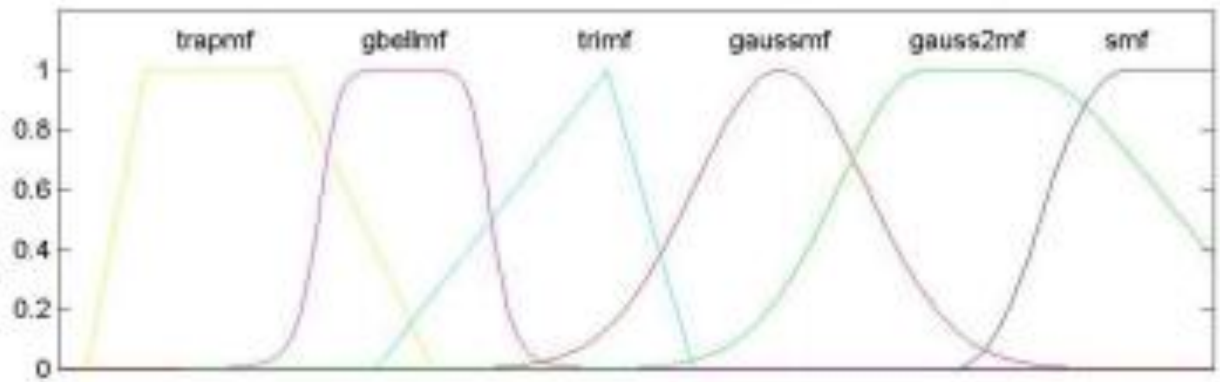
A fuzzy set with only one member with the maximum membership

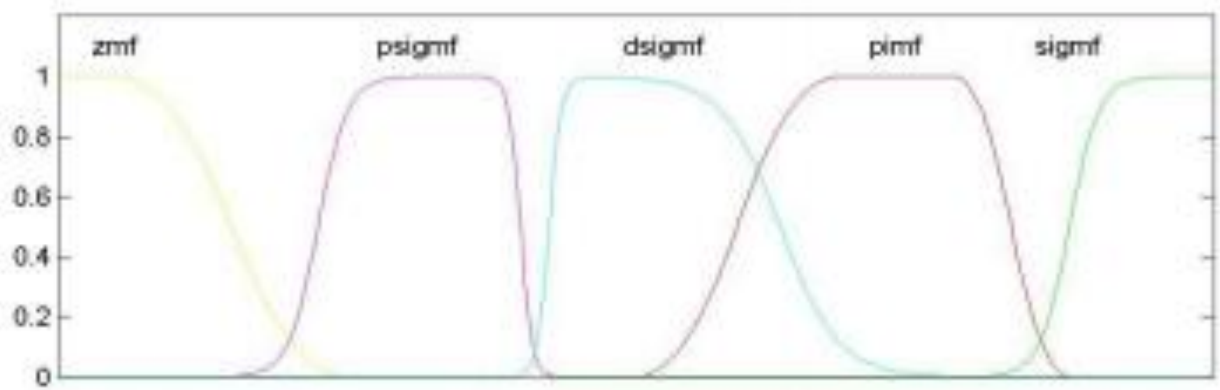


A fuzzy set with a set of members with the maximum membership

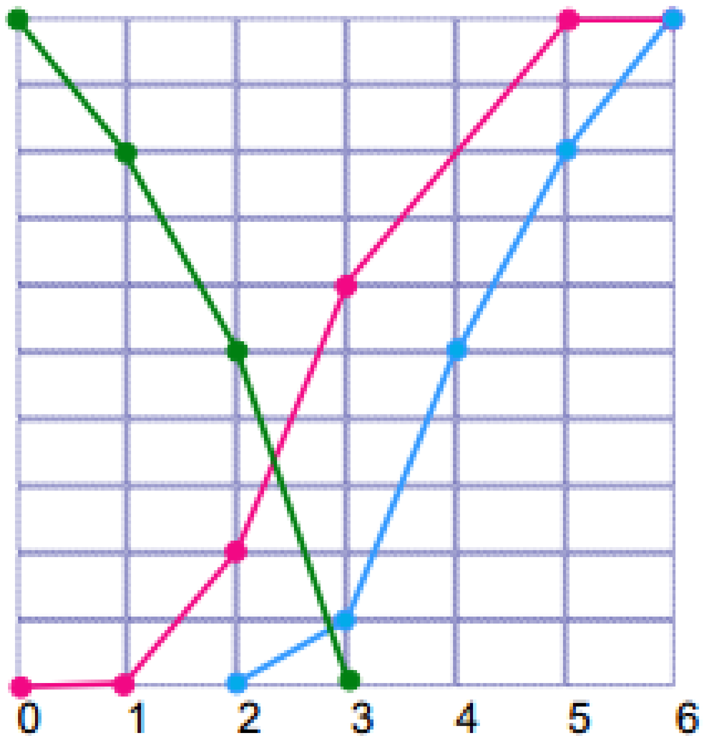


Some variations





Fuzzy sets on ordinal scales



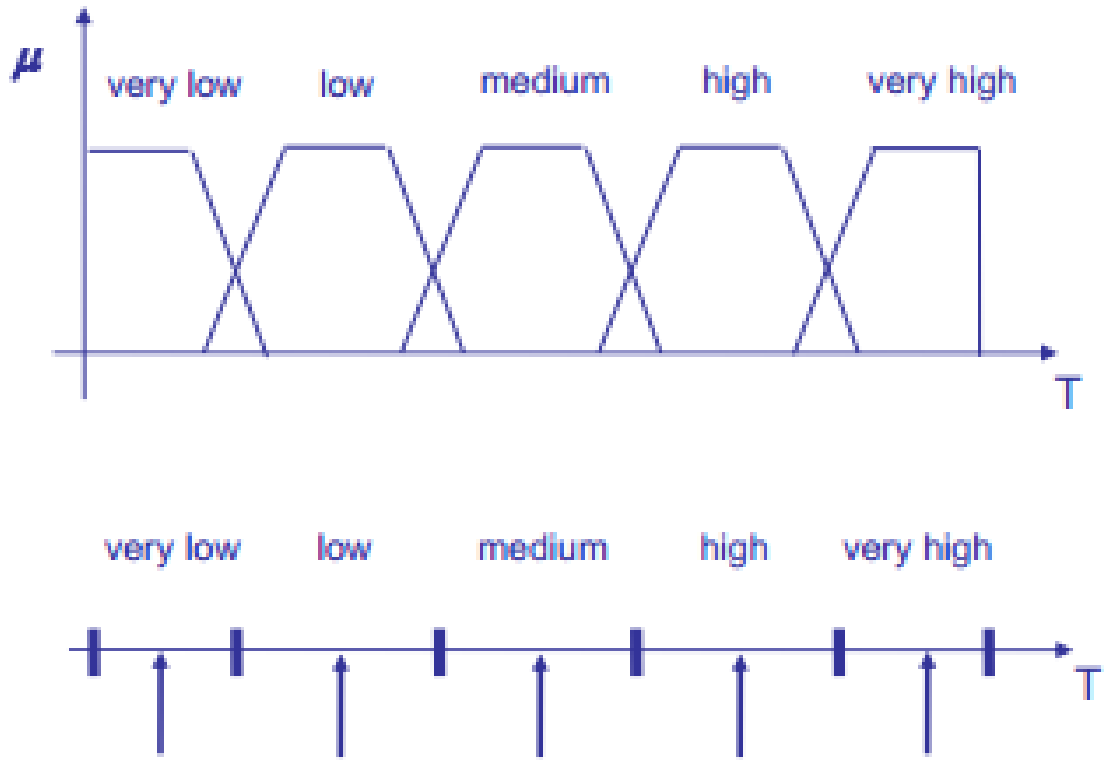
0 - no education
1 - elementary school
2 - high school
3 - two year college
4 - bachelor's degree
5 - master's degree
6 - doctoral degree

poorly educated
highly educated
very highly educated

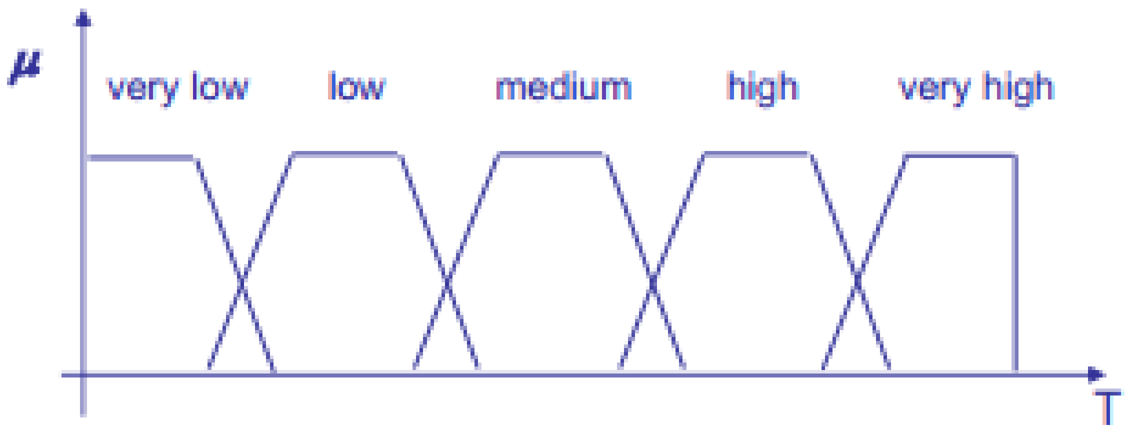
Fuzzy sets and intervals

Smoother transition

in labeling a value



Fuzzy sets covering the universe of discourse



Each fuzzy set is a *granule*

Properties of a frame of cognition

Coverage

Each element of the universe of discourse is assigned to at least a granule with membership > 0

Unimodality **of fuzzy sets**

There is a unique set of values for each granule with maximum membership

Fuzzy partition**:**

for each value of the universe of discourse the sum of membership degrees to the corresponding granules is 1

Robustness

Let's consider a punctual error as the sum of the errors of interpretation by fuzzy sets due to imprecise measurements, noise, ...

$$e(\hat{a}) = |\mu_1(\hat{a}) - \mu_1(a')| + \ldots + |\mu_n(\hat{a}) - \mu_n(a')|$$

and the integral error, as the integral of e(a) over the range of a

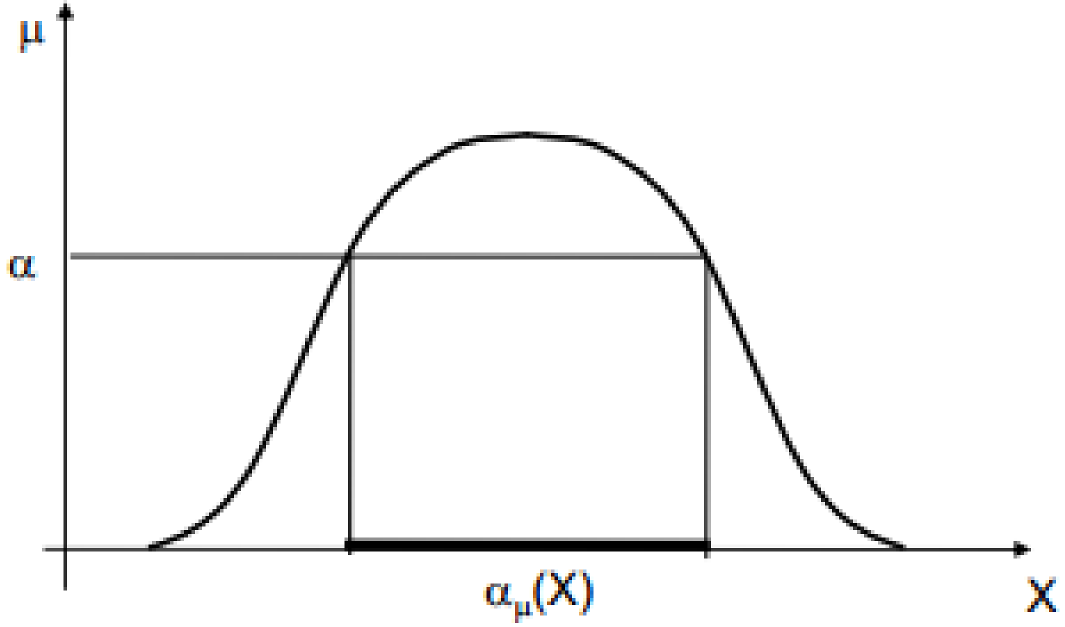
$$e_i = \int e(a)\, da$$

It can be demonstrated that the integral error of a fuzzy partition is smaller than that of a boolean partition, and that it is minimum w.r.t. any other frame of cognition.

α-cuts

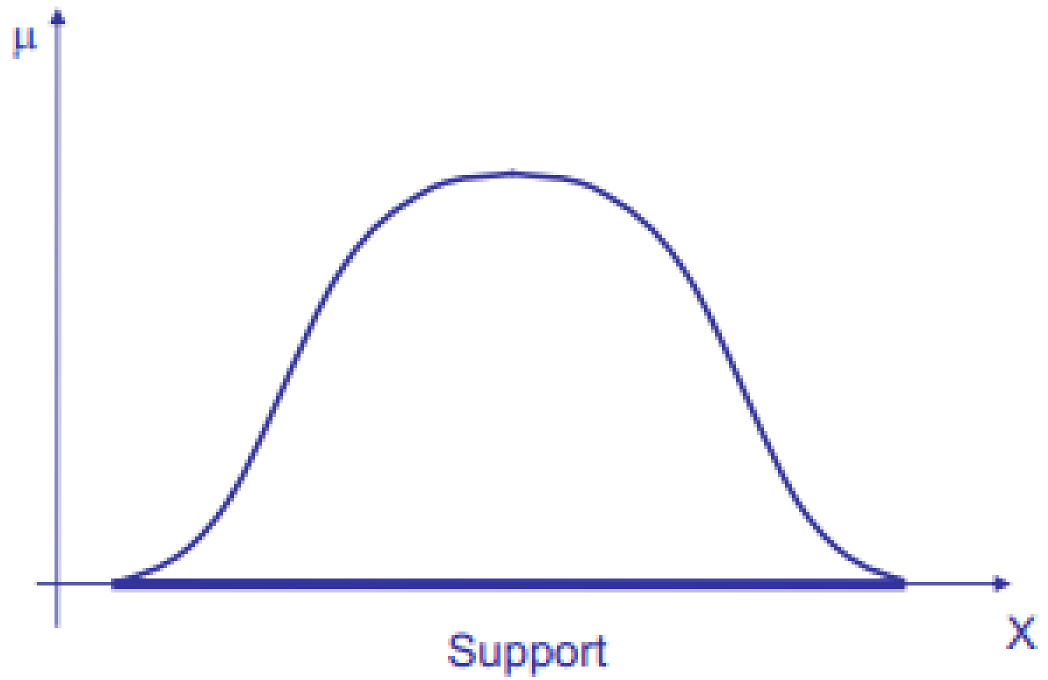
The α-cut of a fuzzy set is the crisp set of the values of x such that $\mu(X) \geq \alpha$

$$\alpha_\mu(X) = \{x \mid \mu(x) \geq \alpha\}$$



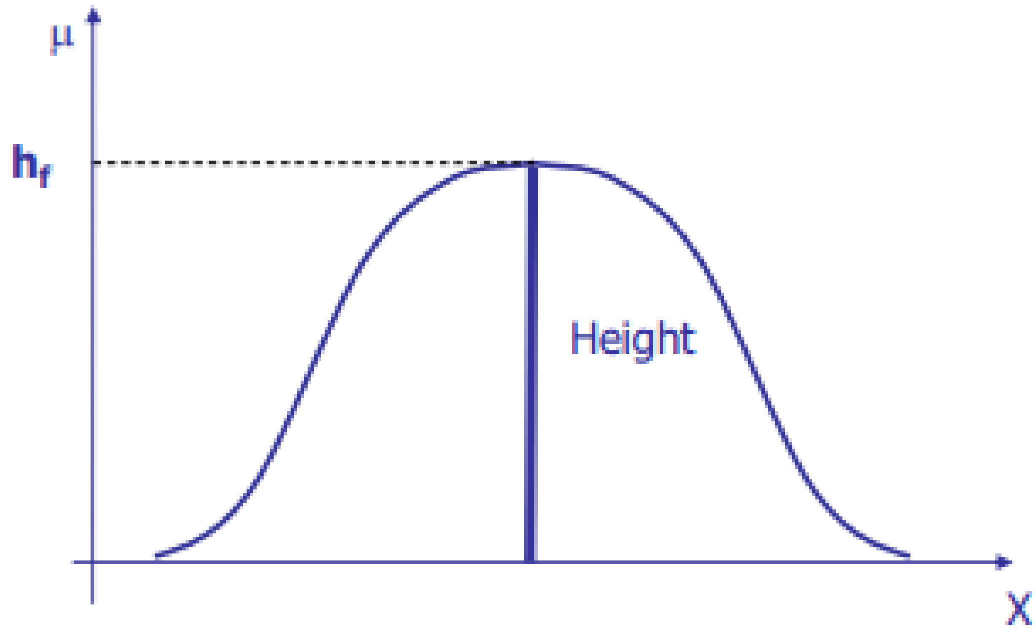
Support of a fuzzy set

The crisp set of values x of X such that $\mu_f(x) > 0$ is the support of the fuzzy set f on the universe X



Height of a fuzzy set

The height *h(A)* of a fuzzy set A on the universe X is the highest membership degree of an element of X to the fuzzy set



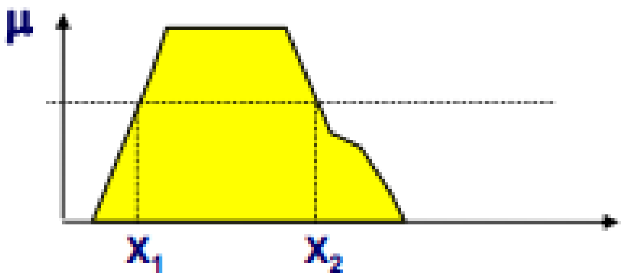
A fuzzy set f is *normal* iff $h_f(x)=1$

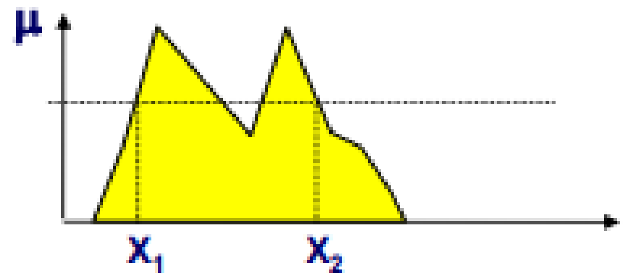
Convex fuzzy sets

A fuzzy set is *convex* iff

$$\mu(\lambda x_1 + (1-\lambda)\, x_2) \geq \min\,[\mu(x_1), \mu(x_2)]$$

for any x_1, x_2 in $\Re$ and any λ belonging to [0,1]

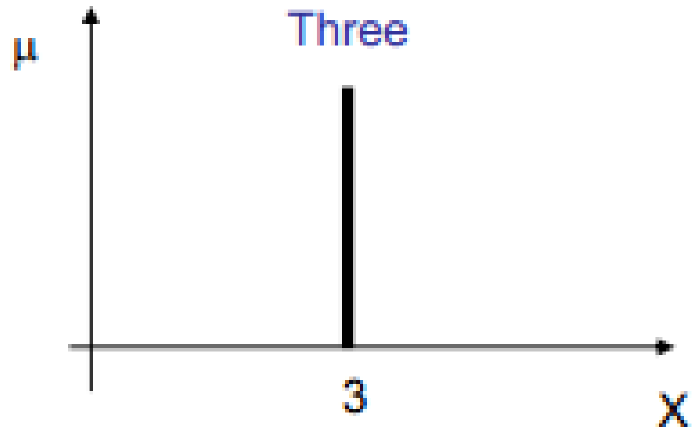




"Strange" MFs

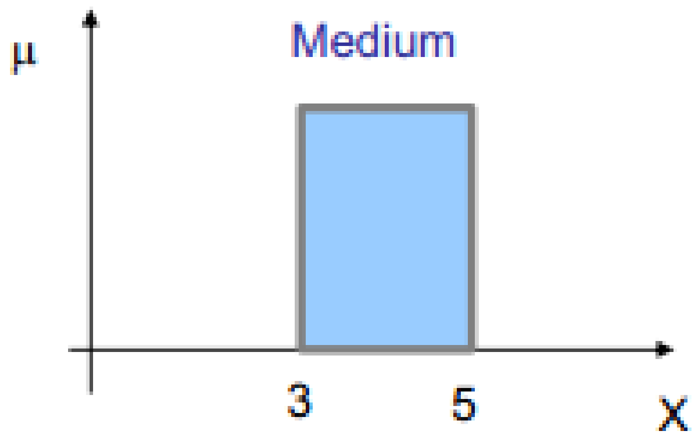
Singleton:

a fuzzy set with one member



Interval:

a fuzzy set whose members have all membership = 1



Operations on Fuzzy Sets

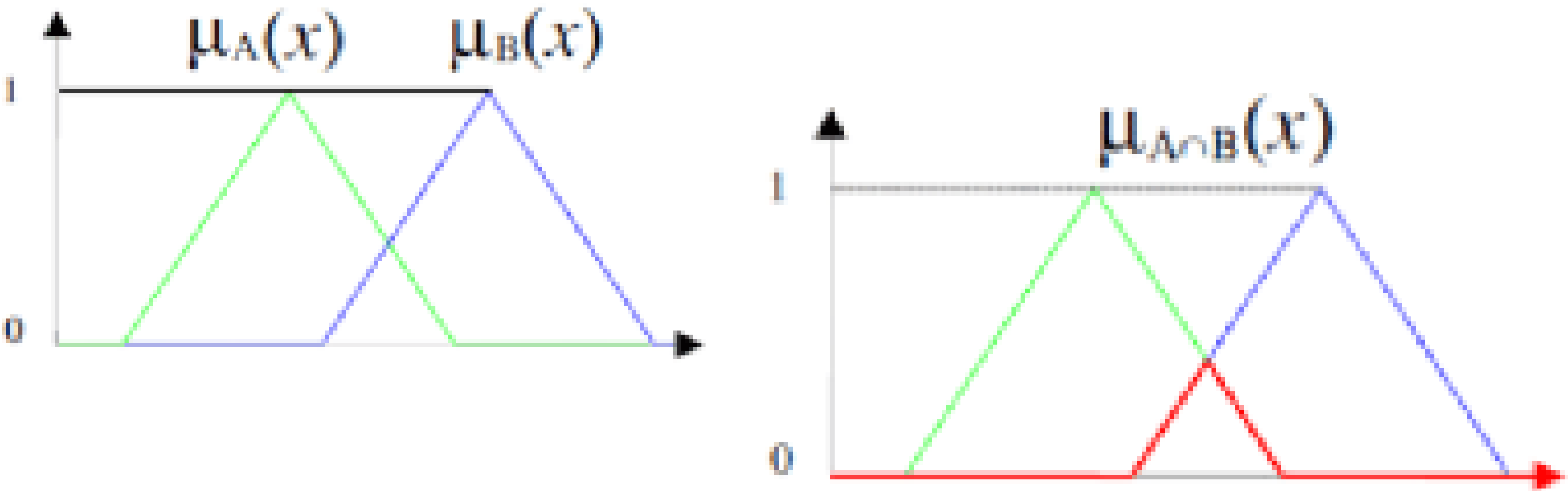
The main feature of Operations on Fuzzy sets is that unlike conventional sets, operation on fuzzy sets are usually described with reference to membership function.

Common Operation defined on Fuzzy sets are:

1. Intersection or minimum function
2. Union or maximum function
3. Fuzzy Complementation

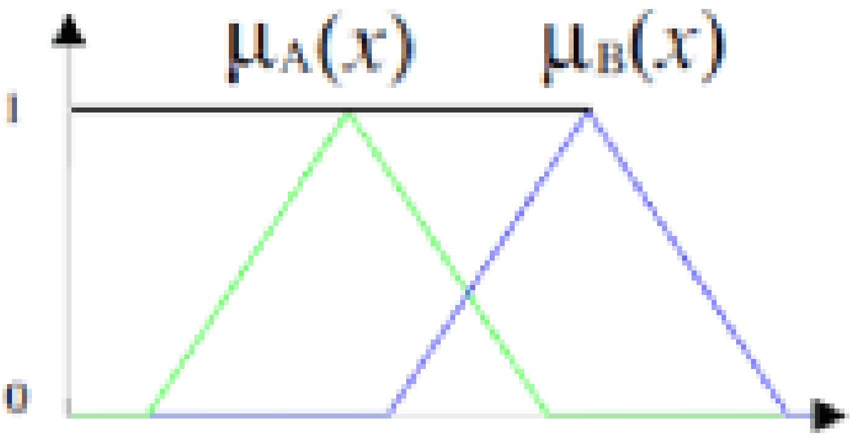
Fuzzy Intersection

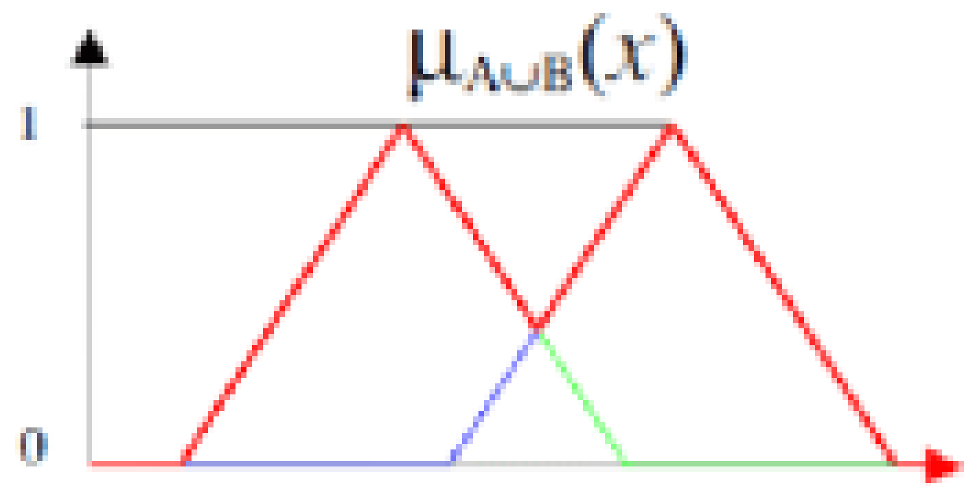
Intersection : $\mu_{(A \cap B)}(x) = \min(\mu_{(A)}(x), \mu_{(B)}(x))$



Fuzzy Union

Union : $\mu_{(A\cap B)}(x) = \max(\mu_{(A)}(x), \mu_{(B)}(x))$

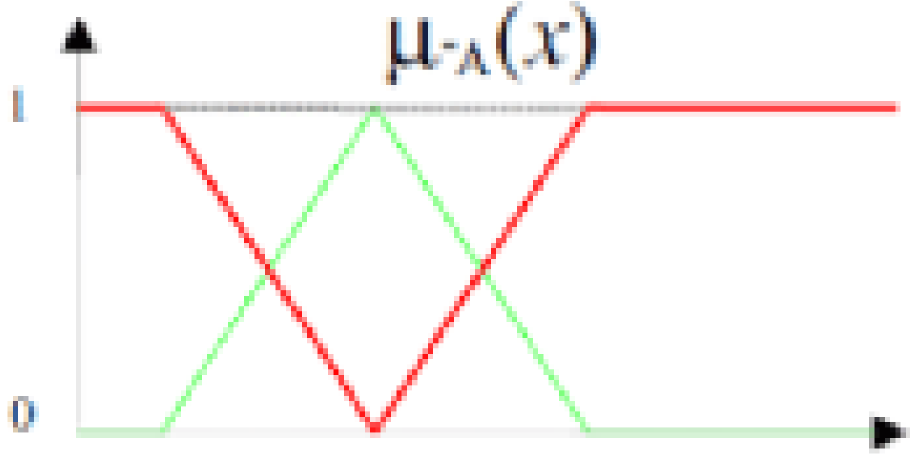




Fuzzy Complement

Complement

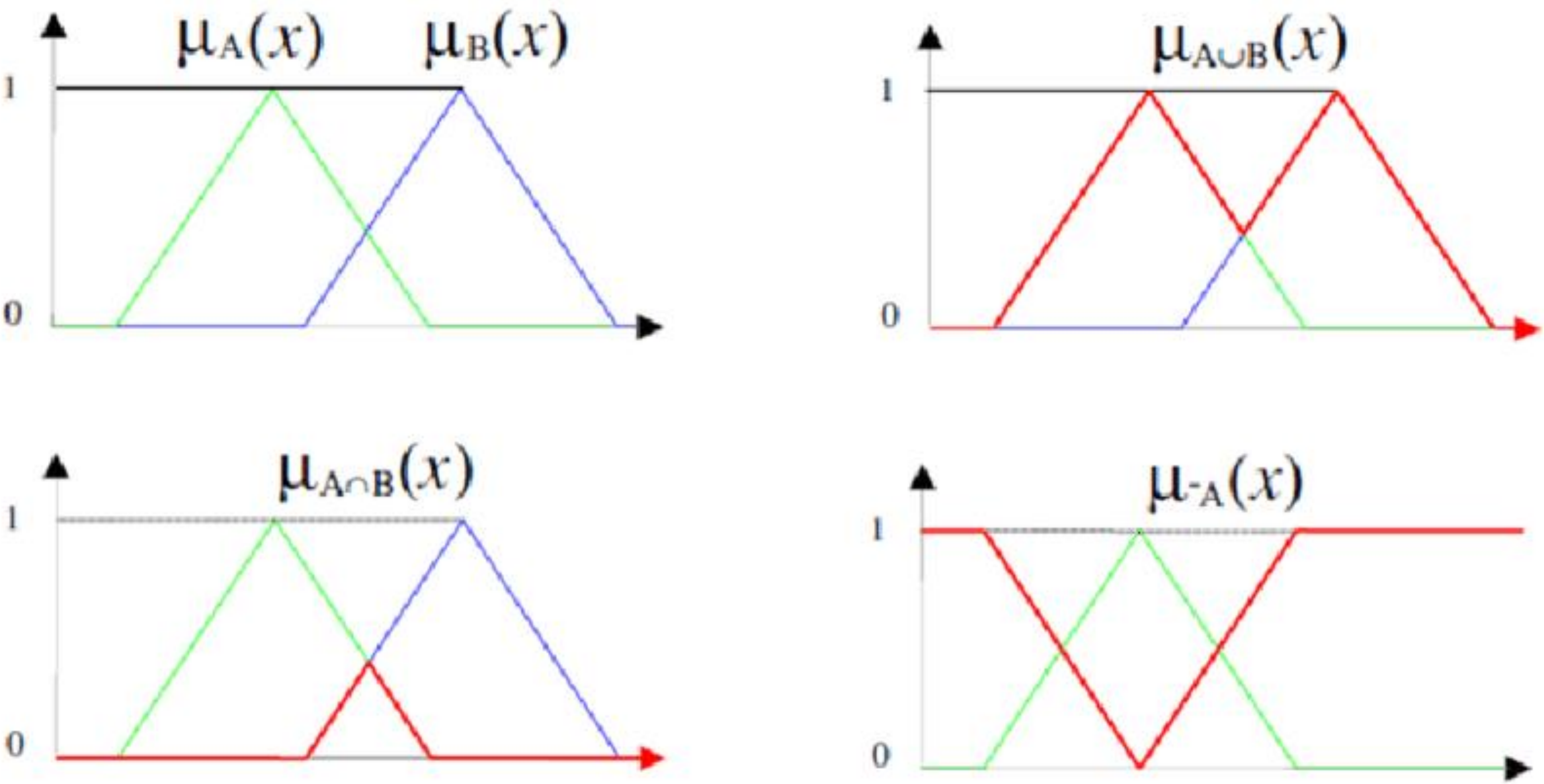
$$\mu_{\bar{\underset{\sim}{A}}}(x) = 1 - \mu_{\underset{\sim}{A}}(x).$$



Union : $\mu_{(A\cap B)}(x) = \max(\mu_{(A)}(x), \mu_{(B)}(x))$

Intersection : $\mu_{(A\cap B)}(x) = \min(\mu_{(A)}(x), \mu_{(B)}(x))$

Complement : $\mu_{(\bar{A})}(x) = 1 - \mu_{(A)}(x)$



Examples

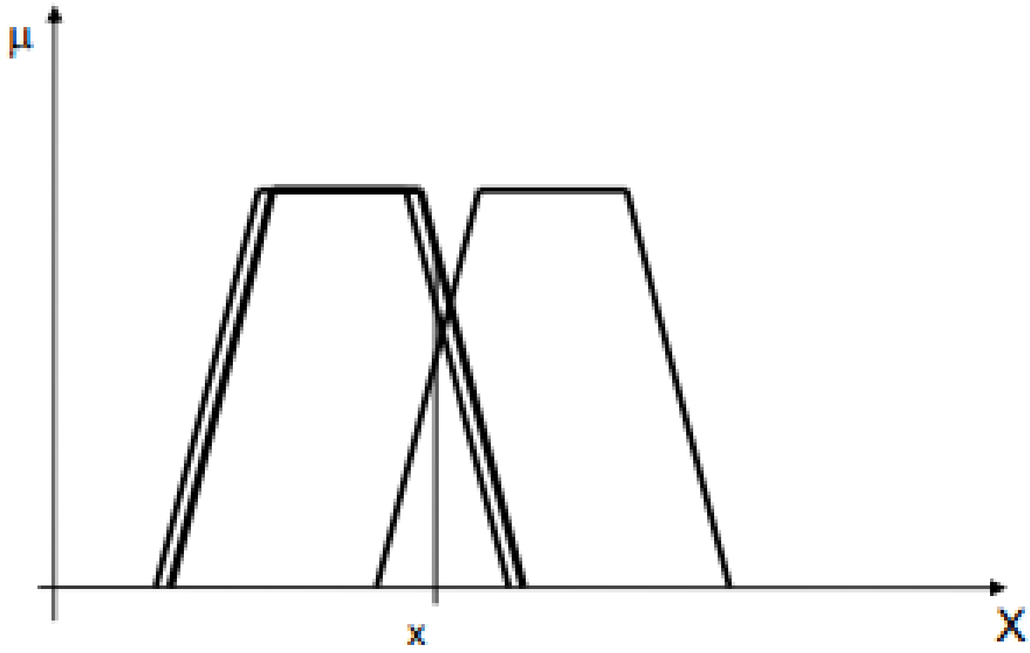
Consider two fuzzy sets A and B find Complement, Union and Intersection.

$$\underset{\sim}{A} = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.6}{4} + \frac{0.2}{5} + \frac{0.6}{6} \right\};$$

$$\underset{\sim}{B} = \left\{ \frac{0.5}{2} + \frac{0.8}{3} + \frac{0.4}{4} + \frac{0.7}{5} + \frac{0.3}{6} \right\}.$$

Fundamental property of standard operators

Using the standard operators the maximum error is the one we have on the operand's MFs



Complement

$$c : [0,1] \to [0,1]$$

$$c(\mu_A(x)) = \mu_{\neg A}(x)$$

Axioms:

1. **c(0)=1; c(1)=0 (*boundary conditions*)**
2. **For all a and b in [0,1], if a < b then c(a) ≥ c(b) (*monotonicity*)**
3. **c is a *continuous function***
4. **c is *involutive*, i.e., c(c(a))=a for all a in [0,1]**

Intersection and T-norms

$$\mu_{A \cap B}(x) = i[\mu_A(x), \mu_B(x)]$$

Axioms:

1. $i[a, 1]=a$ (*boundary conditions*)
2. $d \geq b$ implies $i(a,d) \geq i(a,b)$ (*monotonicity*)
3. $i(b,a) = i(a,b)$ (*commutativity*)
4. $i(i(a,b),d) = i(a,i(b,d))$ (*associativity*)
5. i is continuous
6. $a \geq i(a,a)$ (*sub-idempotency*)
7. $a_1 < a_2$ and $b_1 < b_2$ implies that $i(a_1,b_1) < i(a_2,b_2)$ (*strict monotonicity*)

T-norms: examples

Given the axioms, the intersection operator **i** is defined as a T-norm

A parametric formulation of a class of T-norms:

$$T_\alpha(a,b) = \frac{ab}{max[a,b,\alpha]}$$

for α=1 we have *ab*

for α=0 we have *min(a, b)*

Other T-norms:

$$T_1(a,b) = max(0, a+b-1)$$

$$T_{2.5}(a,b) = \frac{ab}{a+b-ab}$$

Union and T-conorms (S-norms)

$$\mu_{A \cup B}(x) = u[\mu_A(x), \mu_B(x)]$$

Axioms:

1. u[a, 0]=a (*boundary conditions*)

2. b ≤ d implies u(a,b) ≤ u(a,d) (*monotonicity*)

3. u(a,b) = u(b,a) (*commutativity*)

4. u(a,u(b,d)) = u(u(a,b),d) (*associativity*)

5. u is continuous

6. u(a,a) ≥ a (*super-idempotency*)

7. $a_1 < a_2$ and $b_1 < b_2$ implies that $u(a_1,b_1) < u(a_2,b_2)$ (strict *monotonicity*)

S-norms: examples

The most common ones:

$$S_3(a,b) = max(a,b)$$

$$S_+(a,b) = a + b - ab$$

Other S-norms:

$$S(a,b) = min(1, a^p, b^p)^{1/p} \qquad p \geq 1$$

$$S_1(a,b) = min(1, a + b)$$

Aggregation

$$\mu_A(x) = h[\mu_{A1}(x), \ldots, \mu_{An}(x)]$$

Axioms:

1. h[0,..., 0]=0, h[1,..., 1]=1 (*boundary conditions*)

2. *monotonicity*

3. h is continuous

4. h(a,..,a) = a (*idempotency*)

5. *simmetricity*

Properties of aggregation

$$\min(a_1, \dots, a_n) \le h(a_1, \dots, a_n) \le \max(a_1, \dots, a_n)$$

Example of aggregation operator: generalized average

$$h(a_1, \dots, a_n) = (a_1^{\alpha} + \dots + a_n^{\alpha})^{1/\alpha} / n$$

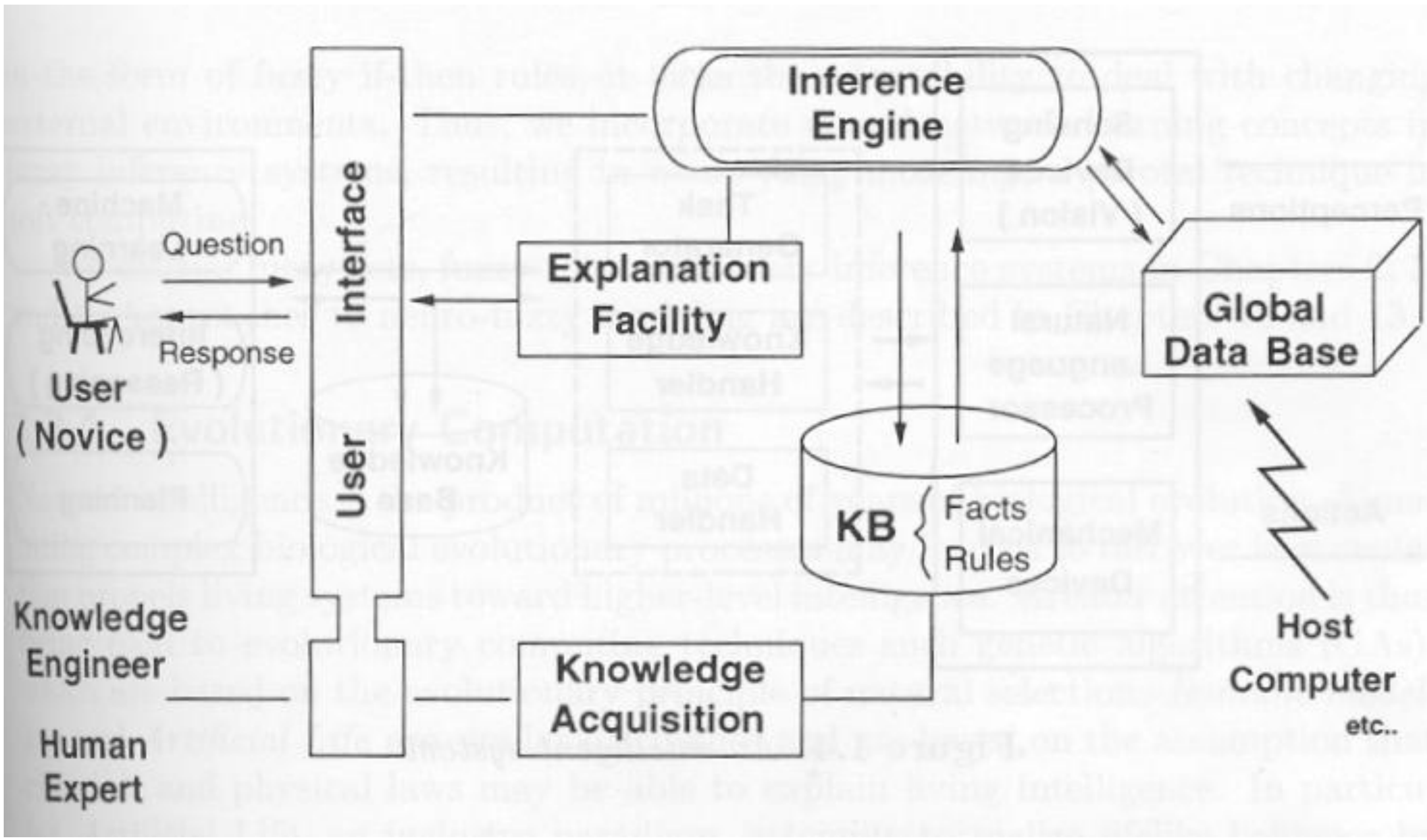
What to remember from these slides?

- **Definition of fuzzy set**
- **Definition of membership function, support, height, α-cut, convex fuzzy set**
- **Main operators**

What is an expert system?

An **expert system** is software that uses a knowledge base of human expertise for problem solving, or to clarify uncertainties where normally one or more human experts would need to be consulted

Expert system



Building blocks of expert system

- Knowledge base: factual knowledge and heuristic knowledge
- Knowledge representation: in the form of rules
- Problem solving model: forward chaining or backward chaining
- Knowledge base: knowledge gained by an individual user

Note:-

- Knowledge engineering:- building an expert system
- Knowledge engineers:- practitioners.

Applications of expert system

1. Diagnosis and Troubleshooting of Devices and Systems of All Kinds
2. **Planning and Scheduling**
3. Configuration of Manufactured Objects from Subassemblies
4. **Financial Decision Making**
5. Knowledge Publishing
6. **Design and Manufacturing**

Questions